Title: CREDIT LENS AI

Team Name: NEXBIT

Team Members:

1)M Sinchana

2)T A Lahari

3)S Punith

Theme: FIN-TECH

PROBLEM STATEMENT



A good credit score doesn't always guarantee that a person has the financial ability to repay a loan, while people with low or no credit history, such as students, first-time borrowers, part-timers, and freelancers, may be financially capable but struggle to prove their creditworthiness. The current system doesn't always reflect a person's complete financial situation. Therefore, there is a need for a universal system that tracks and analyses a person's income, expenses, existing loans, spending habits, and overall financial behaviour to assess their actual repayment capacity, regardless of their credit score or employment type.

Solutions And Existing Market Gaps

Existing Solutions:

- Traditional credit bureaus
- Bank and lender assessments
- Alternative credit assessment
- Account aggregator framework

TransUnion CIBIL



Market Gaps:

- Credit score limitations
- Limited credit history
- Irregular income
- Fragmented financial information
- Limited explainability

Proposed Solution and Uniqueness

Proposed solution: Credit Lens AI Credit Lens AI is an AI-powered financial assessment platform that analyses a person's overall financial situation to help estimate their loan repayment capacity. It considers both people with established credit histories and those with limited or irregular financial records.

Uniqueness:

1.Beyond Credit Scores Evaluates a person's overall financial situation instead of depending solely on traditional credit scores.

2. Universal Financial Assessment Designed for students, first-time borrowers, salaried employees, freelancers, gig workers, and small business owners.

3. Irregular Income Analysis Considers income fluctuations to understand whether someone with an unstable monthly income can manage loan repayments

4. AI-Powered Repayment Simulation Simulates repayment affordability under different situations, such as reduced income, increased expenses, or unexpected financial commitments.

5. Explainable AI Explains the factors influencing a financial assessment in simple language instead of presenting only a score

6. Personalized Financial Reliability Report Generates a report highlighting income stability, expenses, existing debts, savings, and estimated repayment affordability**.**

CreditLens AI
Technical Architecture & Tech Stack

TECHNICAL ARCHITECTURE

User Interface (React)
- Web / Mobile App
- Forms & Dashboards
- Charts & Visualisations
- Responsive design

→ **API Layer** (FastAPI)
- Handles requests
- Data validation
- Authentication & security

→ **Backend Server** (Python)
- Business logic
- Data processing
- Model integration

→ **Database** (SQLite)
- Stores user data
- Financial records
- Session data

→ **AI Explanation Engine** (Optional LLM API)
- Generates explanations
- Personalised insights
- Financial recommendations

→ **Output Reports**
- Dashboard (web)
- PDF Report
- Download / Share

CORE PROCESSING & DATA FLOW

Data Collection
- User input
- Income & expenses
- Loan details
- Savings & debts

→ **Data Preprocessing**
- Clean & validate data
- Handle missing values
- Categorise transactions

→ **Analysis & Simulation**
- Income stability
- Expense analysis
- Repayment simulation
- Scenario testing

→ **AI Explanation**
- Simple explanations
- Risk analysis
- Personalised suggestions

→ **Final Report**
- Financial summary
- Graphs & comparisons
- Downloadable PDF

DATA SOURCES (Optional)

User Provided
(Personal & Employment Info)

Bank Statements
(Manual Upload / Future Integration)

Financial Institutions
(With Consent)

External Data
(Defaults, Market Data)

Prototype: manual input / synthetic data; no bank login or credit bureau integration required.

TECH STACK

Frontend
React · Tailwind CSS

Backend
Python · FastAPI

Database
SQLite

Data Processing
Pandas · NumPy

AI / LLM
OpenAI API (Optional)

Visualisation
Plotly

PDF Generation
ReportLab

Tools
Git · VS Code · (Optional) Docker

Feasibility, Impact & Future Scope

1.Feasibility

Technical: Uses React, Python, Fast API, Pandas, and SQLite for financial analysis and data management.

Cost-effective: Built using open-source technologies with minimal infrastructure requirements.

Practical: Analyses income, expenses, savings, and existing debts to simulate repayment affordability.

2.Impact

Financial Inclusion: Supports individuals with limited credit histories and irregular incomes.

Responsible Borrowing: Helps users understand repayment affordability before taking loans.

Transparency: Provides clear, explainable insights into financial strengths and potential risks.

Better Assessment: Offers a broader view of financial capacity beyond credit scores alone.

3.Future Scope

Automated Analysis: Integrate consent-based bank statement analysis.

Predictive AI: Develop validated models to estimate repayment risk using appropriate historical data.

Secure Integration: Explore partnerships with financial institutions and authorised financial-data platforms.

Personalisation: Provide tailored financial insights and advanced repayment simulations.